

Student Resource Guide

Generative AI for Beginners

Section 4: Image Generation AI | Lecture 4.5: Prompting for Image Generation

Learning Objectives

- Identify the key structural elements of an effective image generation prompt
- Apply systematic prompt iteration techniques to improve generation quality
- Use negative prompts, attention weighting, and style modifiers effectively
- Diagnose common prompt failures and apply targeted corrections
- Develop a repeatable prompt engineering workflow for production use cases

Introduction

Prompt engineering for image generation has become a distinct professional skill. While early image generators required highly technical prompt syntax, modern models respond to natural language with remarkable flexibility. However, understanding the mechanics behind why certain phrases work—and others don't—transforms prompt writing from intuitive guessing into a systematic, reproducible practice. For practitioners deploying image generation in commercial contexts, this reliability is what separates a useful tool from an unpredictable one.

Effective image prompts share characteristics with good technical specifications: they are unambiguous, prioritised, and complete. The mental model of 'directing a very literal artist who has seen every image on the internet but cannot read your mind' is practically useful. Everything you want in the image must be stated; everything you don't want should appear in the negative prompt. Context that seems obvious to a human viewer must be made explicit.

This guide extends the lecture's core concepts with practical frameworks, style vocabulary, and a systematic debugging workflow. By the end, you should be able to construct effective prompts on first attempt for common use cases, and diagnose and correct failures efficiently when they occur.

Key Concepts

Prompt Anatomy and Element Priority

Definition	A well-structured image prompt contains: subject (who/what), subject attributes (descriptive properties), action/state (what
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	is happening), environment (where), lighting (illumination character), artistic style, compositional framing, and quality modifiers. CLIP text encoders give higher weight to earlier tokens, making element ordering significant.
Business Context	Structured prompts produce consistent, repeatable results across generation runs—essential for maintaining visual brand standards in commercial workflows. Unstructured prompts produce high variance, requiring multiple retries and human selection.
AI Practitioner relevance	Practitioners maintaining image generation pipelines should maintain a prompt library with documented structures for each use case. Treating prompts as versioned artefacts (like code) enables systematic quality improvement and debugging.
Real-World Example	A structured e-commerce prompt: '[Subject: white leather handbag], [Attributes: minimalist design, gold hardware], [Environment: marble surface, studio], [Lighting: soft diffused light, slight reflection], [Style: professional product photography], [Quality: highly detailed, 8K]' — each element is explicit and front-loaded by importance.

Style Vocabulary and Artistic References

Definition	Style modifiers are keywords that activate specific visual aesthetic patterns learned from CLIP's training data. These include artistic media (oil painting, watercolour, digital art), movement names (impressionism, cyberpunk, brutalism), photographer names (Annie Leibovitz, Ansel Adams), and rendering quality terms (photorealistic, concept art, illustration).
Business Context	Style vocabulary enables rapid production of content in specific brand aesthetics without hiring specialists in each style. A brand can define a style prompt once and produce unlimited consistent assets in that aesthetic.
AI Practitioner relevance	Practitioners should build a domain-specific style vocabulary by systematically testing which style terms produce the

	desired aesthetic for their use case. Not all style terms in the internet's vocabulary are equally well-represented in CLIP's training data.
Real-World Example	Testing 'corporate photography style' vs. 'professional business photography, Getty Images editorial style, natural lighting' on the same subject reveals significantly different outputs—the second, more specific formulation produces more consistent, commercially usable results.
Key Risks	Style references based on living artists may produce outputs that closely mimic their distinctive style, raising ethical and legal questions about intellectual property. Prefer generic style descriptors for commercial work unless the artist's work is clearly in the public domain or licensed.

Negative Prompts

Definition	Text that is provided to the model alongside the positive prompt to specify what should NOT appear in the output. Mechanically, the negative prompt is used in classifier-free guidance to steer generation away from unwanted visual features.
Business Context	Negative prompts are essential for production quality control. Without them, common artifacts (blurry images, anatomically incorrect hands, watermarks, low-contrast areas) appear regularly. A well-maintained negative prompt library reduces manual review and retries significantly.
AI Practitioner relevance	Practitioners should maintain tiered negative prompt sets: a universal baseline (always-on), use-case-specific additions (e.g., for portraits: 'extra limbs, asymmetrical eyes'), and content-policy negatives (anything prohibited by the deployment context).
Real-World Example	Standard baseline negative prompt: 'blurry, out of focus, low quality, watermark, signature, duplicate, mutated, disfigured, extra limbs, poorly drawn hands, missing fingers, low resolution, jpeg artifacts, pixelated'. This single string reduces

	artifact rate substantially across all use cases.
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Attention Weighting and Prompt Blending

Definition	Syntax features in some image generation UIs that allow practitioners to increase (weight > 1.0) or decrease (weight < 1.0) the attention given to specific tokens during cross-attention conditioning. Implemented as (word:1.3) for emphasis or (word:0.7) for de-emphasis in many tools.
Business Context	Attention weighting enables fine-grained visual control without changing the overall prompt. Emphasising a specific colour, texture, or compositional element helps resolve cases where the model gives insufficient weight to a critical visual attribute.
AI Practitioner relevance	Use weighting conservatively—values above 1.5 often produce oversaturation or visual distortion. The practical range for effective emphasis is 1.1–1.4; for de-emphasis, 0.6–0.9. Always test weighted prompts at multiple seeds to confirm the effect is stable.
Real-World Example	For a prompt 'a woman in (red:1.3) dress standing in a green field', the red dress receives 30% more attention weight—increasing the probability that the dress is correctly rendered red rather than conflated with the surrounding green environment.

Real-World Applications

Prompt engineering skills apply across every commercial image generation use case. The following scenarios demonstrate how structured prompting produces reliable, production-ready outputs.

Brand-Consistent Marketing Asset Generation: Define a brand prompt template that encodes your brand's visual language: colour palette, photographic style, subject treatment, and quality standards. Apply this template consistently across all generation runs to produce brand-consistent assets without photoshoots. Version-control the template and review quarterly.

A/B Creative Variant Generation: Use prompt ablation to systematically generate visual variants for A/B testing. Start with a base prompt and modify one attribute at a time (background colour, model age, product angle, lighting style). This produces clean experimental variants where the changed attribute is the only independent variable.

Accessibility Image Generation: Use inclusive prompt language to produce diverse representation in generated imagery. Explicitly specify age, ethnicity, body type, and ability status where not specifying would default to the model's training distribution (typically skewed toward specific demographic representations from internet data).

Technical and Instructional Illustration: For technical content (infographic-style diagrams, process illustrations, instructional step images), use style prompts like 'flat design illustration, clean lines, minimal shadows, corporate infographic style'. Combine with negative prompts excluding photorealistic and painterly aesthetics to avoid inappropriate style mixing.

Best Practices

Build and maintain a prompt library. Document prompts that work well for your use case in a shared repository. Include notes on what each element contributes, which negative prompts are included, and the model version used. Treat prompts as reusable intellectual property.

Use the PASE framework: People, Action, Setting, Emotion. This structure covers the most common failure points: who is in the image (People), what they are doing (Action), where they are (Setting), and what the mood or feeling should be (Emotion). Complete this before adding style and quality modifiers.

Test prompts at multiple seeds before declaring them 'solved'. A prompt that looks great at one seed may fail at others. Test at 3–5 random seeds to assess prompt robustness—only prompts that succeed across seeds are reliable for production.

Calibrate negative prompts per model version. Negative prompts that are effective for SD1.5 may be unnecessary or ineffective for SDXL. Test your standard negative prompt set when upgrading model versions.

Document failed prompts as much as successful ones. Understanding why certain phrasings fail—concept confusion, overloaded attention, token budget exceeded—is as valuable as knowing what works. Maintain a failure log with hypotheses for each case.

Common Mistakes to Avoid

Mistake: Describing what you don't want in the positive prompt instead of using a negative prompt.

Correct approach: Phrases like 'no blurriness' or 'without watermarks' in the positive prompt do not reliably suppress those features—CLIP does not handle negation reliably. Place all unwanted features in the dedicated negative prompt field where CFG mechanics correctly handle them.

Mistake: Using ambiguous quantity and spatial language that CLIP interprets inconsistently.

Correct approach: CLIP has weak spatial and quantitative reasoning. 'Three apples on the left' frequently produces two or four apples, or apples on the right. Use compositional ControlNet or inpainting for precise spatial control; do not rely on prompt language alone for exact positioning.

Mistake: Copying prompts from prompt-sharing websites without understanding each element.

Correct approach: Shared prompts are often tuned for specific model versions, resolutions, and samplers. Applying them in a different context produces unpredictable results. Understand each element, then adapt to your specific configuration rather than copying wholesale.

Practitioner Tool: Image Generation Prompt Diagnostic Workflow

Type: Prompt Debugging Checklist

When an image generation output does not match expectations, work through this checklist in order to identify and fix the root cause.

- Verify the subject is stated explicitly at the start of the positive prompt
- Count prompt tokens — ensure critical content is within the first 60 tokens
- Check for ambiguous or contradictory descriptors (e.g., 'realistic and abstract simultaneously')
- Confirm style modifiers are specific (named artist/movement) not generic ('nice', 'good')
- Review negative prompt — ensure unwanted features are negated here, not in positive prompt
- Test at 3 different seeds to distinguish prompt failure from random seed failure
- Try increasing CFG scale by 2 points to test if prompt adherence improves
- Test removing the last 20% of the positive prompt to check for truncation effects
- If spatial/compositional issues persist, consider ControlNet or inpainting instead of prompt-only solution
- Compare output quality with a simpler version of the prompt to identify overloading

Note: Work through this checklist top-to-bottom for systematic diagnosis. Most prompt failures are resolved by steps 1–5.

Knowledge Check

Question 1: A practitioner writes 'no blurry backgrounds' in the positive prompt but continues to get blurry backgrounds. What is the correct fix?

- A. Increase CFG scale to 20 to force prompt compliance
- **B. Add 'blurry' and 'out of focus' to the negative prompt field instead**
- C. Remove all lighting descriptors from the positive prompt
- D. Use a higher resolution to avoid blurring automatically

Correct Answer: B. Add 'blurry' and 'out of focus' to the negative prompt field instead

Explanation: CLIP handles negation poorly in positive prompts. The negative prompt field is mechanically implemented through CFG to correctly steer generation away from specified visual features.

Question 2: A brand team generates product images using a well-crafted prompt and gets excellent results at seed 42, but inconsistent results at other seeds. What does this indicate?

- **A. The prompt is overfit to a specific random seed and is not robustly constructed**
- B. The model needs more inference steps
- C. The VAE decoder is outputting different quality levels
- D. The model has reached its token limit

Correct Answer: A. The prompt is overfit to a specific random seed and is not robustly constructed

Explanation: A production-ready prompt should produce acceptable results across multiple seeds. Seed-dependent quality means the prompt is relying on the random initialisation to 'accidentally' solve compositional or attribute problems rather than the prompt providing sufficient guidance.

Question 3: A designer needs to generate an image with exactly three specific objects in precise spatial positions. After multiple attempts, the prompt consistently fails to place objects correctly. What is the most appropriate technical solution?

- A. Add exact pixel coordinates to the prompt
- B. Increase attention weight on each object to 2.0
- **C. Use ControlNet or inpainting for precise spatial control instead of prompt language**
- D. Generate 100 images and select the one with correct positioning

Correct Answer: C. Use ControlNet or inpainting for precise spatial control instead of prompt language

Explanation: CLIP-conditioned diffusion models have weak spatial and quantitative reasoning. Precise positional control requires architectural tools (ControlNet, which conditions on a spatial reference image) rather than text prompting alone.

Reflection Questions

Take time to consider the following questions before the next session:

1. Prompt engineering skills are increasingly valuable yet non-standardised across the industry. How should organisations document and transfer prompt knowledge as models and tools evolve?
2. The most effective prompts are often counterintuitively specific. How does this challenge the assumption that AI tools reduce the need for domain expertise?
3. Negative prompts work by leveraging the CFG mechanism to steer away from unwanted outputs. Are there types of visual problems that negative prompts cannot solve, and what alternative techniques address those?
4. As image generation models improve at understanding natural language, will highly specific technical prompt engineering remain necessary, or will it converge toward simple natural language description?
5. What ethical considerations arise when practitioners deliberately use prompt engineering to produce images of specific people, demographics, or sensitive scenarios?

Key Takeaways

- Effective image prompts have a structure: subject → attributes → environment → lighting → style → quality modifiers, with critical elements front-loaded for CLIP's token weighting.
- Negative prompts are mechanically implemented through CFG to exclude visual features—negation in positive prompts is unreliable and should not be used.
- Style vocabulary (artistic movements, named photographers, rendering terms) is one of the highest-leverage levers for changing output character.
- Attention weighting (word:1.3) allows fine-grained control of which prompt elements dominate—use conservatively in the range 1.1–1.4.
- Test all prompts at multiple seeds before declaring them production-ready; single-seed success indicates seed dependence, not prompt robustness.
- Spatial and quantitative precision requires architectural solutions (ControlNet, inpainting) rather than text prompting alone.

Next Actions

Complete at least two of the following before the next lecture:

6. Take one image generation prompt you currently use and rewrite it following the structured anatomy (subject, attributes, environment, lighting, style, quality). Test both versions at the same seed and compare.
7. Build a baseline negative prompt string for your most common use case and test it against 10 generations—measure the reduction in artifact rate.
8. Experiment with attention weighting: take a prompt where one element is often rendered incorrectly and add (element:1.2) weighting. Test whether this resolves the issue at 5 different seeds.
9. Create a prompt template document for one specific use case (product photography, portrait, landscape) with the prompt structure, standard negatives, recommended CFG, and tested style terms.